

## Question 19 – Stage 1

Primrose chooses a prime number between 2 and 98 inclusive at random and calls it  $P$ .

Eve chooses an even number between 2 and 98 inclusive at random and calls it  $E$ .

Simultaneously they recite successive terms in their own arithmetic sequence:

Primrose has first term  $P$  and common difference  $E$ .

Eve has first term  $1000E$  and common difference  $-P$ .

What is the probability (to 3 sig.figs) that, at some stage, they say the same number at the same time?

### Working

$$\begin{aligned} \text{First term is } n = 0 \\ y_1 &= P + nE \\ y_2 &= 1000E - nP \\ P + nE &= 1000E - nP \\ nP + nE &= 1000E - P \\ n(P + E) &= 1000E - P \\ n &= \frac{1000E - P}{P + E} \end{aligned}$$

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In [3]: from itertools import product
from pyTools import roundSF
from sympy import primerange

def isValidN(n): # for n to be a valid index, it must be a +ve int
    if int(n) == n and n >= 0:
        return True
    return False

primes = list(primerange(98))
evens = list(range(2,99,2))

totalCount = 0
possibleCount = 0

for p,e in product(primes,evens):
    totalCount += 1
    possibleN = (1000*e-p)/(p+e)
    if isValidN(possibleN):
        possibleCount += 1

probability = possibleCount / totalCount
print(f"Probability: {roundSF(probability,3)}")
```

Probability: 0.0596